

Conductive-Proof Building Materials Suite

Cost Analysis per Unit — CSMFAB 010

Category: Aegis Home — Structural **Unit Basis:** complete set: 1 door frame + 10 fastener sets **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
Pultruded FRP door frame sections (3m set)	12 kg	\$22.00/kg	\$264.00
3Y-TZP ZrO2 fastener set (M10×50, 20pc)	0.4 kg	\$120.00/kg	\$48.00
MAX Phase Ti3AlC2 joint inserts (8pc)	0.3 kg	\$400.00/kg	\$120.00
BFRP lattice flooring tiles (1m²)	4 kg	\$22.00/kg	\$88.00
YInMn Blue glaze on frame	0.2 kg	\$75.00/kg	\$15.00
PEEK CF40 structural clips (bag of 50)	0.5 kg	\$280.00/kg	\$140.00
BOM Subtotal			\$675.00

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$675.00	\$553.50	\$438.75
Direct Labor	\$280.00	\$180.00	\$120.00
Overhead (15% of labor)	\$42.00	\$27.00	\$18.00
Quality Control & Testing	\$33.60	\$14.40	\$6.00
Packaging & Logistics	\$25.00	\$20.00	\$15.00

Tooling & Equipment Notes: Pultrusion die: \$8,000 NRE. ZrO2 fastener injection mold: \$12,000 NRE.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$1,480	\$2,368 – \$3,256	35-55%
Small Batch (100-500/yr)	\$980	\$1,470 – \$1,960	33-50%
Production Volume (5K-50K/yr)	\$680	\$951 – \$1,224	28-45%

Market Reference: \$800-1,200 (construction premium market; no RF zoning issues)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$121 – \$236
- **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

End of Conductive-Proof Building Materials Suite Cost Analysis | CSMFAB 010 | June 2026